

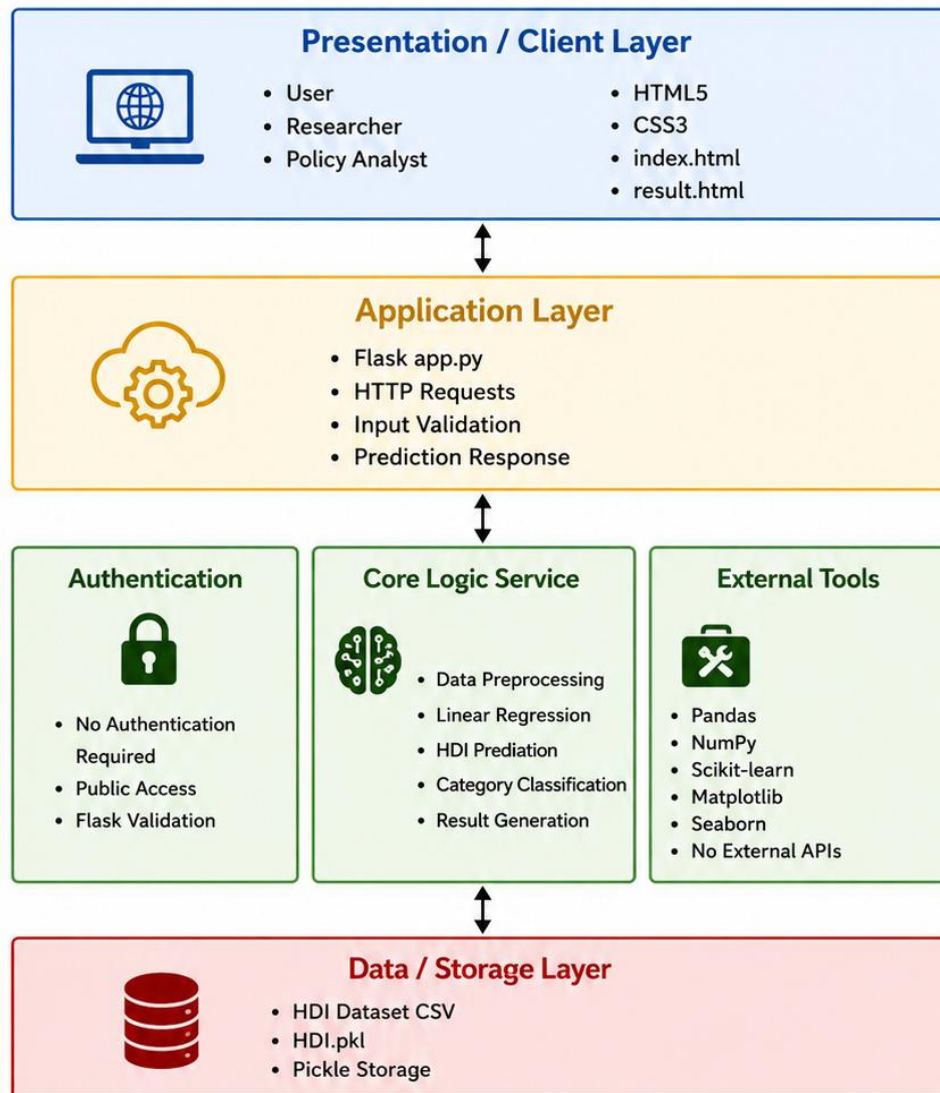
Solution Architecture

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Solution Architecture Diagram – Human Development Index (HDI) Prediction System



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides the user interface for inputting socio-economic indicators and displaying predicted HDI results.	HTML5, CSS3
Application Layer	Handles HTTP requests, validates inputs, and returns prediction response using Flask.	Flask (app.py), HTTP
Authentication	Ensures secure and open access to the system through validation without requiring user authentication.	Flask Validation
Core Logic Service	Performs data preprocessing, builds the model, predicts HDI, classifies category, and generates results.	Pandas, NumPy, Scikit-learn
External Tools	Provides supporting libraries for data manipulation, modeling, and visualization without external API dependencies.	Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
Data / Storage Layer	Stores the dataset and trained model artifact required for HDI prediction.	CSV, Pickle (.pkl)

